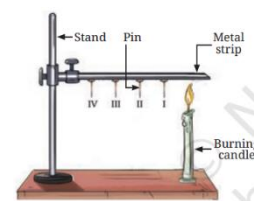


## Chapter – 7: Heat Transfer in Nature

- Pema, Palden – reside – Gangtok
- Cold winter evening – sitting around fireplace
- Palden – shares experience – visiting Kerala – winter vacation
- Compared to Gangtok – Kerala winters – warm, humid
- Both – curious – some places – cold – others – quite hot
- Their grandfather – retired science teacher – explains – Kerala – closer to equator – also long coastline – result – warmer, more humid weather
- Palden replies – learnt – Grade 6 – Earth – src. of light, heat – Sun – around equator – climate – generally hot
- Along with talking – Pema obs. – grandmother – cooking *thukpa* – large metal pan
- Pema asks – why cooking utensils – made of metal?
- Palden – replies – chapter – metals and non-metals – such materials – good conductors – heat

### Conduction of Heat

- Activity –
  - Take – strip of metal – aluminium, iron – 15 cm long
  - Attach – 4 pins – help of wax – arranged – equal dist.
  - Secure – strip to stand – label pins – I, II, III, IV
  - Heat – end of strip – away from stand – using candle
  - **Predict** – order – pins fall – **record**
- Obs. – pin closest – candle – falls first
- **Infer** – heat – travels along strip – approaches pin – wax melts – pin falls
- Here – transfer of heat – hot end to colder end
- Process – heat transfer – hotter part to colder part – **conduction**
- This process – particles – heat up – passes heat – neighbour – BUT – particles – do not move
- Materials – allow heat – pass through – **good conductors**
- Metals – good conductors – cooking utensils – made of metals
- Solids – heat transfer – mainly through conduction
- Use strip – material – wood, glass – instead of metal – above activity – pins do not fall
- Materials – glass, wood – do not allow heat – pass through – **poor conductors** (insulators)
- Clay, porcelain – poor conductors – tea, coffee – inside them – stays hot longer
- List materials – around us – **classify** – good, poor conductors
- During winters – prefer woollen clothes – keep warm
- Woollen fabric – traps air – poor conductor – reduces heat flow – body to surroundings
- Air trapped b/w layers of clothes – poor conductor – keeps warm
- Houses – very cold, hot climate – use concept – heat transfer – keep cool, warm
- Houses – outer walls – hollow bricks – keeps warm – winters – keeps cool – summers
- Pema – draws attention – rising smoke – why always up?



### Convection

- Activity –
  - Take – 2 identical paper cups



- Hang them – equal lengths – thread – 2 ends – wooden stick
- Adjust position – cups – stick – horizontal
- Place – burning candle – below 1 cup – obs. – record
- Cup – above candle – rises up
- Air – around candle – heats up – air inside cup – warms up – expands – occupies more space – result – lighten up – rises up
- Experience – expansion of air on heating – use – partially inflated balloon under sun
- Air in balloon – gets heated – expands – balloon gets larger
- Must obs. – incense stick (*agarbatti*) – burnt – smoke rises
- Smoke – mixture – hot gases, tiny solid particles – released – something burns
- Smoke – lighter than – air – rises up
- Activity –
  - Take – 500 mL beaker – fill  $\frac{1}{2}$  – water
  - Use straw – place grain – potassium permanganate – centre – beaker's base
  - Place – candle – right below the centre – base of beaker
  - Obs. – movement – colour streak – water
  - Supply heat – streak of colour – moves up – comes down – sides
- Why – such behaviour?
- Water – bottom of beaker – heats up – becomes hot – expands – lightens up – rises
- This cycle – continues – until – entire volume – heats up
- This case – entire volume – heats up – movement of particles
- This process – heat transfer – **convection**
- Convection – reason – movement of colour streak – inside beaker
- **Conclude** – water – like air – heats up – convection – heat transfer – takes place – actual movement – particles – liquids, gases

### Land and sea breeze

- Palden – shares experience – visit beach – Kerala – winter vacation
- During day – sand, soil – near beach – hotter than water
- BUT – at night – sand, soil – cooler than water
- Pema replies – diff. obj. – heat, cool differently
- Activity –
  - Take – 2 identical bowls
  - One bowl – soil – other bowl – water – both halfway
  - Fix – lab thermometer – each bowl – make sure – bulbs – immersed – soil, water – do not touch – bottom, sides
  - Place – setup – sunlight
  - Measure – temp. – soil, water – every 5 min. – record
  - Study – rise in temp. – soil, water
- 20 min. later – temp. – soil – rises more than water – indicates – soil – heats faster
- Bring setup – indoors – let it cool – 20 min.
- Obs. – soil – cools – faster than water
- People – coastal areas – experience – interesting phenomenon – heating, cooling – land, water – diff. rates
- Land – heats faster – warm air – above land – rises – cooler air – moves – sea to land – called – **sea breeze**
- Hot places – sea breeze – relieves people – windows – placed – facing sea

- Night time – process reverses
- Absence of sunlight – land – cools down faster – result – air above sea – warmer – rises up – cooler air – moves – land to sea – **land breeze**
- People – near seashore – experience – dir. – wind reverses – day, night

## Radiation

- Pema, Palden – sitting near fireplace – felt warm
- Grandfather explains – heat transfer – this case – directly – fire to us – **radiation**
- Heat of Sun – reaches us – this process
- Heat transfer – radiation – no medium req.
- All obj. – radiate heat – utensils – away from flame – cools down
- Hot utensil – cools down – radiating heat – surroundings
- Many examples – daily life – obs. – conduction, convection, radiation – all together
- Consider – water – heated up
- **Identify** – various ways – pan, water – heats up – warmth – we feel – around flame, hot pan
- Heat transfer – flame to utensil – conduction
- Water in utensil – heats up – convection
- Warmth – feel around – flame, hot utensil – radiation
- From act., ex., discussions – we find – 3 processes – heat transfers –
  - Conduction – heating – particles – transfer heat – no movement
  - Convection – heat transfer – movement of particles
  - NOTE – conduction, convection – medium req. – heat transfer
  - Radiation – heat transfer – not material medium req.
- Learnt – Grade 6 – textbook – *curiosity* – Sun – main src. of heat – Earth
- Seen parents – drying clothes – clothesline – home
- Wet clothes – dry faster – sunny day – heat of Sun – makes evaporation faster
- Heat from Sun – imp. role – evaporation – clothes drying – water bodies – lakes, oceans

## Water Cycle

- Also learnt – Grade 6 – *curiosity* – water – 3 states
  - Liquid – fills – oceans, rivers, lakes
  - Solid – forms snow, ice sheets, glaciers – mountains, polar regions
  - Gas – water vapour – earth's atm.
- During summers – some snow – melts – Sun's radiation – flows down – rivers, oceans
- Melted ice – replenished – fresh snow – winters
- Water – oceans, rivers, lakes – heats up – Sun – result – evaporates – water vapour
- Evaporation – also – trees, plants – transpiration
- Water vapour – rises – cools down, condenses – form clouds
- Clouds – rain, snow, hail – this process – **precipitation**
- Continuous movement – water – upward – vapours – downward – precipitation – pass through – soil, rocks, plants – return – water bodies – **water cycle**
- Water cycle – helps – redistribute, replenish – water – rivers, lakes, oceans – also – conserve – total amt.
- Rainwater – falls on earth – flows – ponds, lakes, rivers, oceans, seeps into grnd.

- Seen people – drawing water – wells, handpumps – this water – seeped into grnd.

### Seepage of water beneath the Earth

- Activity –
  - Take – 3 transparent, used – plastic bottle – 1 L
  - Cut – middle – make hole – cap – each bottle
  - Keep – inverted – pour – 1<sup>st</sup> – clay – 2<sup>nd</sup> – sand – 3<sup>rd</sup> – gravel
  - Place – 3 identical beakers – below bottles
  - Add – 200 mL water – each bottle
  - Predict – amt. of water – flows out – each bottle
  - Collect water – flows out – each bottle – 10 min. – compare
- Obs. – water flow – fastest – gravel – slowest – clay
- Space b/w particles – wider in gravels – compare to sand, clay – water – seeps faster
- This process – surface water – seeping through grnd. – **infiltration**
- Water – infiltrate faster – space b/w particles – wider, open, interconnected
- Water – seeps through – stored – pore spaces, openings – sediments, rocks – **groundwater**
- Underground layers – sediments, rocks – store water – called **aquifers**
- This water – we extract – digging, boring wells into aquifers
- This water – few meters to few 100 meters – depend – location
- Groundwater – not unlimited – growing water req. – increasing population – excessive extraction
- Decreased vegetation – increased concrete surface – limited infiltration
- Result – groundwater – depleted – address this – rainwater harvesting, recharge pits – replenish groundwater
- Water cycle – ensure – groundwater src. – recharged
- Water scarcity – life difficult – people – diff. ways – conserve water
- Ladakh – innovative ways – ice stupas